

Sarang Shastry

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EDUCATION

PhD in Computer Science BITS Pilan KK Birla Campus, Goa *Aug 2026 – Present*

- Pursuing doctoral research in Theoretical Computer Science

Master of Computer Applications Gyan Ganga College of Technology *Graduation: 2023*

- GPA: 8.58

Bachelor of Computer Applications Nachiketa College, Jabalpur *Graduation: 2021*

- Percentage: 61%

RESEARCH EXPERIENCE

Research Internship BITS Goa *May 2026 – Jul 2026*

- Worked on Theoretical Computer Science, with a focus on Interactive Theorem Proving and Cryptography.
- Explored formal methods and proof assistant tools in the context of cryptographic protocols.

Research Collaboration Swansea University *Jan 2026 – Present*

- Working on **Formalizing Polynomial-Time Cryptographic Adversaries Using Rocq (Interactive Theorem Prover)**.
- Modeling adversarial behavior under polynomial-time constraints within a proof assistant.
- Studying foundational links between cryptography, computational complexity, and interactive theorem proving.

TECHNICAL SKILLS

- Proof Assistants: Coq (Rocq), Lean, NuSMV
- Programming: C++, Python, Java, OCaml, Haskell
- Tools: LaTeX, VS Code, Microsoft Office

RESEARCH INTERESTS

- Formal Verification
- Cryptography
- Computational Complexity

TEACHING EXPERIENCE

Faculty Member Divysrijan IAS Academy *2024 – 2025*

- Taught computer science and mentored students for exams.
- Designed assignments and evaluated performance.

Assistant Professor Thevanai Ammal College *2023 – 2024*

- Designed and delivered undergraduate CS courses including Java, Data Structures and Algorithms, Theory of Computation, and Discrete Mathematics.
- Used interactive teaching methods to improve learning.

PROJECTS

Customer Churn Prediction using logistic regression (Python, scikit-learn)

- Developed a machine learning model to predict customer churn using logistic regression on a telecom dataset.
- Performed data pre-processing, feature engineering, and model evaluation with scikit-learn, identifying key churn drivers like tenure and monthly charges.

Sales Analysis Dashboard using Tableau

- Designed an interactive Tableau dashboard for sales performance analysis, integrating data from multiple sources to visualize trends, regional sales, and top products.
- Enabled stakeholders to drill down into KPIs like revenue growth and customer segmentation.

SQL-based data analysis with joins and window functions

- Conducted advanced SQL queries on e-commerce datasets using complex joins, window functions (e.g., ROW_NUMBER, LAG), and aggregations to uncover insights on customer behavior, order patterns, and cohort retention, optimizing queries for large-scale data processing.

WORKSHOPS

Lean for Programmers Indian Institute of Science (IISc), Bangalore

Dec 2025

- Attended intensive workshop at IISc on applying Lean methodology to software development.
- Gained hands-on experience in theorem proving, formal verification, and interactive programming in Lean.
- Explored Lean 4 features for dependently-typed programming and its applications in software verification.